

States of Matter

Melting point: The temperature at which a pure solid turns to a liquid when heated

Freezing point: The temperature at which a pure liquid turns to a solid when cooled

Boiling: The process in which a liquid turns to a gas when heated to the boiling point

Evaporation: The process in which molecules at the surface of the liquid gain enough energy to turn to gas before the boiling point.

Condensation: The process by which gas particles turn to liquid when cooled

Sublimation: The process by which a solid changes directly to a gas without passing through liquid state

Differences between:

Boiling	Evaporation
- Happens at a specific temperature	- Happens at a range of temperature
- Happens throughout the liquid.	- Happens at the surface of the liquid
- Bubbles form	- No bubbles form

Factors affecting evaporation.

1- Temperature: When temperature increases, the kinetic energy of molecules increases, so particles are able to escape the forces of attraction and evaporate faster.

2- Surface area exposed: When increasing the surface area, the number of molecules in contact at the surface of the liquid increases, so evaporation takes place rapidly.

3- Air current above the surface: When air is blown above the surface of the liquid, it will take away the newly formed gaseous particles from the air above the liquid, resulting in an increase of evaporation rate.

The main points of the Kinetic theory:

- 1- All matter is made up of very small particles called atoms
- 2- The volume of the individual particle is considered negligible
- 3- Particles move all the time
- 4- Collisions between particles are elastic. no energy is lost or gained, it can only be transferred.
- 5- The pressure of a gas is produced by the atoms of the gas hitting the walls of the container
- 6- The movement and arrangement of particles is different for the three states of matter.
- 7- The average kinetic energy of a gas is assumed to be directly proportional to the Kelvin temperature

→ The particles in a solid are arranged in a regular lattice structure or pattern. Strong forces of attraction hold the particles close together. They cannot leave their place or move, they can only vibrate in place.

How does melting happen?

When the temperature is raised, particles gain more kinetic energy and vibrate more strongly, so they occupy more space. The solid expands. Eventually, the particles will gain enough energy to break the forces holding the lattice structure, and they can move around; the solid melts.

→ The particles in a liquid are packed closely together in an irregular arrangement. Weak forces of attraction hold the particles together. Particles can slide over each other.

How does evaporation & boiling happen?

When heating, particles gain more kinetic energy, the liquid expands. Some particles will have enough energy to escape from the surface of the liquid; Evaporation. When the temperature reaches the boiling point, the particles have enough energy to break the forces of attraction between them, and they move faster, causing them to separate from each other; the liquid boils.

→ The particles in a gas are arranged totally irregularly. They are spread very far apart compared to liquids and solids. They move randomly.

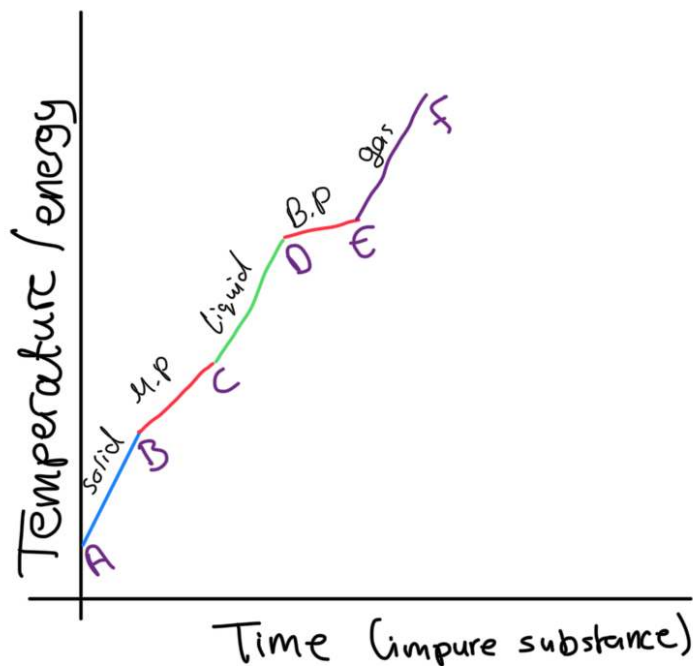
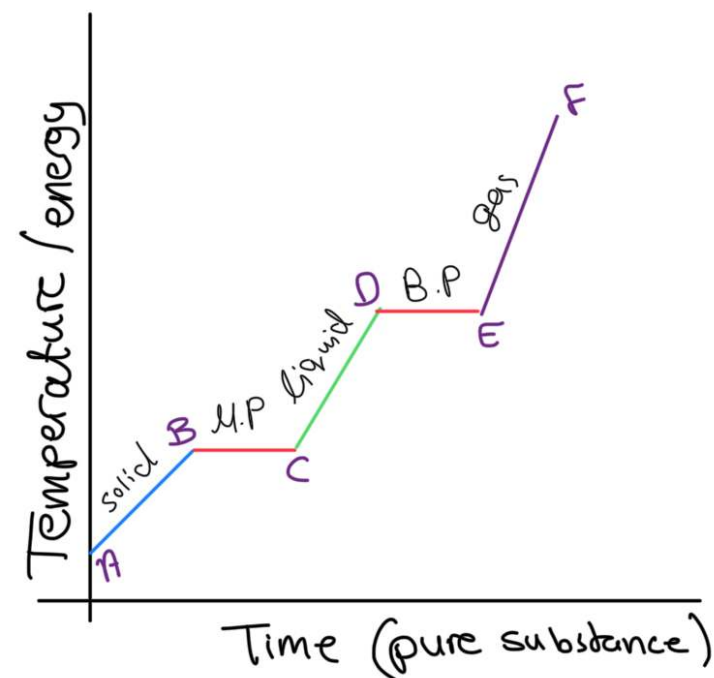
When a substance is impure:

→ Melting point decreases as the presence of impurities make the forces between particles weaker, so less energy is required to break the bonds in solid structure.

→ Boiling point increases as impurity presence make it difficult for liquid particles to change to gas state.

→ Both happen over a range of temperatures because different substances and each one has its own melting and boiling points

Heating curve



A-B: Solid state: As the temperature increases, kinetic energy increases, causing solid particles to vibrate faster → Solid expands.

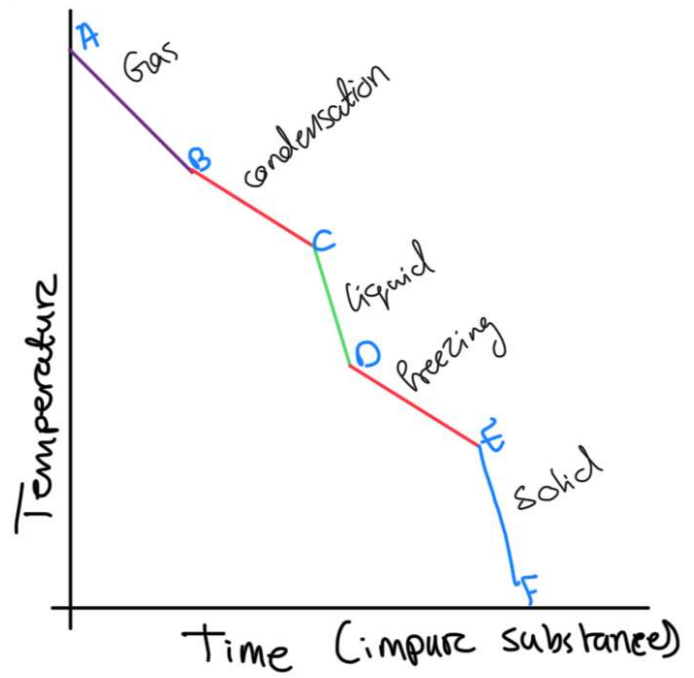
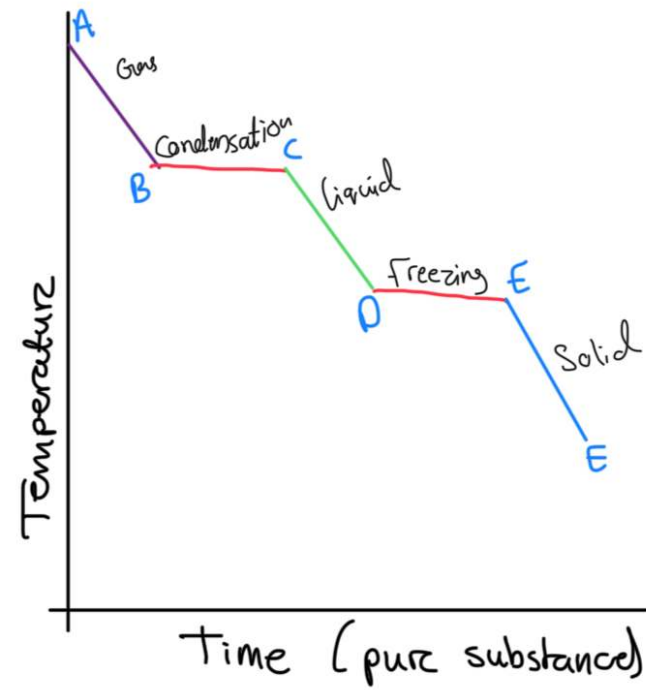
B-C: Melting process: The forces between solid particles will break, changing solid to liquid.

C-D: Liquid state: As the temperature increases, the kinetic energy increases, causing the liquid particles to slide past each other faster. Some of the particles have enough energy to evaporate from the surface of the liquid. Liquid expands.

D-E: Boiling process: The forces between liquid particles will break, turning the liquid to gas.

←→ PE ↑ KE	↑ PE ←→ KE
A-B	B-C: Solid & liquid coexist
C-D	
E-F	D-E: Liquid & gas coexist

Cooling Curve



A-B: Gas state: As the temperature decreases, the kinetic energy decreases, causing particles to move slower and closer to each other, causing bonds to form.

B-C: Condensation: The forces between the gas particles will start to form, turning the gas to liquid

C-D: liquid state: As the temperature decreases, the kinetic energy decreases, the liquid particles will slide past each other slower, and will get closer to each other.

D-E: Freezing: The forces between the liquid particles will start to strengthen, turning the liquid to solid

$\longleftrightarrow$ PE $\uparrow$ KE	$\uparrow$ PE $\longleftrightarrow$ KE
A-B	B-C: gas & liquid coexist
C-D	
E-F	D-E: liquid & solid coexist

Brownian motion: The uncontrolled or erratic movement of particles in a fluid due to their constant collisions with other fast-moving particles. Larger particles can be moved by light, fast moving particles

Diffusion: The movement of Fluids from high concentration to low concentration where particles mix and spread by colliding with other particles.

Factors affecting diffusion:

1- **Temperature:** When temperature increases, the kinetic energy of molecules increases, so particles will move faster and collide with each other more, resulting in faster diffusion.

2- **Concentration:** When more particles are present, it increases the rate of diffusion.

3- **Mass:** The lighter the fluid, the faster the diffusion.

Physical property: The characteristics of the substance

Chemical property: Ability or inability for a substance to react with another substance

Physical change: Maintains the characteristics of the substance

Chemical change: Forms a new substance with new properties